Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/30/2013
PI Name:	Weitz, David Ph.D.		
Project Title:	Physics of Colloids in Space		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02138	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Fluid Physics Rapid Turn Around Flight Opportunity
Start Date:	12/18/2007	**End Date:**	12/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
Contact Email:	william.v.meyer@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date change to 12/31/2013 per NSSC information as of 12/10/2012. In addition, per W. Meyer/GRC, title should be "Physics of Colloids in Space"; also, change in period of performance--should be 12/18/2007-12/17/2012 for first awarded period of performance and supplements; there was no end date change to 3/31/2013 or 10/31/2014 (Ed., 4/21/15) NOTE: End date changed to 10/31/2014 per W. Meyer/R. Sicker at GRC (Ed., 9/24/13) NOTE: End date changed to 10/31/2013 per flight information (Ed., 12/5/12) NOTE: title change to "Binary Colloidal Alloy Test -6" and change in end date to 3/31/2013, per W. Meyer/GRC (Ed., 9/8/2011)		
Key Personnel Changes/Previous PI:			

COI Name (Institution):	Yodh, Arjun (University of Pennsylvania) Chaikin, Paul (New York University) Lynch, Matthew (Proctor & Gamble Company) Hollingsworth, Andrew (New York University) Lu, Peter (Harvard University) Frisken, Barbara (Simon Fraser University) Bailey, Arthur (Simon Fraser University)
Grant/Contract No.:	NNX08AE09G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The BCAT-6 and BCAT-7 investigations build on a heritage of (3 ISS and 2 Shuttle/Mir) flight experiments and they are packed with important scientific and commercial investigations. This effort will enable us to continue to address fundamental questions in colloidal engineering that impact product shelf life and answer how concentrated systems of particles of select sizes and shapes cause order to naturally arise out of disorder when gravity is removed. The baseline proposal is to deliver BCAT-6 for Multi-Purpose Logistics Module (MPLM) stowage on ULF-5 presently planned for 7/29/10. BCAT-7 is planned for late stowage on ULF-5. We propose to conduct four parallel sets of sample monitoring and imaging experiments (20 samples in all): (1) Crystallization of (NYU-provided) ellipsoidal colloidal particles and PS-DNA particles that should form crystals. The PS-DNA tagged particles will be set to crystallize at room (ISS cabin) temperature and be annealed as the ISS cabin temperature cycles a degree or two each side of this whenever the International Space Station (ISS) orbits the earth and is hidden from the sun when it goes behind the earth. PS-DNA also forms the foundational blocks for making future self-replicating structures, (2) Equilibration of (UPenn-provided) graphene disks with liquid crystal properties that should be visible between crossed-polarizers as they approach equilibrium after initially being homogenized (looking at a new generation of highly sensitive detectors and the foundations for zero-resistance substrates based on carbon electronics), (3) Phase separation of colloidal (P&G- and Harvard-provided) samples of commercial interest in a multi-billion dollar industry where the physics underlying shelf-life is masked by gravitation, and (4) the Competition between phase separation and crystallization (with SFU providing samples); this is the cause of defects that are often seen during the extrusion of plastics and a better understanding of the underlying physics should result from additional data points in this investigation. The objectives of these experiments examine different conditions that result in entropically-driven colloidal crystallization, melting, self-organization, and phase separation of systems. The long-duration low-gravity environment provided on the International Space Station (ISS) allows such structures to form and persist for sufficient time to establish the evolution toward equilibrium through time sequenced images taken with the EarthKAM system on the ISS or with the intervalometer of the new camera (until the EarthKAM software is updated to communicate with it). These experiments cannot be done on Earth because of problems caused by sedimentation, convection, and jamming. Being able to track the time evolution allows scientists to see the underlying physics, which is of particular relevance to product manufacturers for the pharmaceutical, food, and cleaning industries (in terms of shelf life and mechanical and thermodynamic properties). Of particular note is to have the first low-gravity flight of self-annealing PS-DNA tagged particles, ellipsoidal colloidal particles, and 2-D graphene carbon sheets in equilibrium. The fundamental questions outlined above can be addressed using two Binary Colloid Alloy Test (BCAT) flight hardware modules that already exist; and the answers provided by these experiments will lay the foundations for later scientific investigations proposed by these investigators for the Advanced Colloids Experiment (ACE) with the Light Microscopy Module (LMM) on the ISS. Funding this proposal accomplishes two objectives. First the direct scientific results will be important on their own and guide the choice of which classes of questions can be resolved in microgravity while telling us what needs to be experimentally tweaked now to get things to work with the LMM (i.e., particle size, polydispersity, concentration, etc.); and second, this work will form a successful bridge for keeping the science teams of existing and future students alive and productive while working with our experienced scientists (PIs) to ensure that these endeavors are optimized and focused while we continue the development of and await the future deployment of the fully functional Light Microscopy Module on the ISS in 2013 or 2014.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have further improved the chemistry and imaging techniques that are directly related to a host of ground-based experiments. In particular, we have developed new GPU-based high-speed image analysis techniques, which we have published previously (FY2011). We are the process of making this technology available for other researchers who may benefit from this new technology.
Task Progress:	We have made significant progress toward a number of goals associated with the colloids effort in microgravity in 2013. We have collected new substantial data working with ISS-based astronauts as part of the Binary Colloid Alloy Test (BCAT) series of experiments, including our samples from the BCAT-3, BCAT4, BCAT-5, and BCAT-6 experiments, essentially concluding data collection from these sets of experiments that have spanned the past ten years. Thanks to the skill of the onboard astronauts and leadership of our ground-based teams at NASA, we have collected data of unprecedented quality and quantity, and are now continuing our analysis and concomitant ground-based experiments. For ACE-M2, we have formulated, characterized, and delivered flight samples for launch, which we have delivered to NASA Glenn Research Center's principal contractor, ZIN Technologies, in cooperation with NASA and ZIN scientists and engineers. We have provided expertise and relevant data on samples, their loading and imaging, and design and specification of related hardware included as parts of the light microscopy module (LMM). In June 2013, we presented a week-long short course on the physics of colloids and advanced imaging, at NASA Glenn Research Center, which brought together scientists, engineers, and leaders from several NASA centers and contractors, for an intensive workshop focusing on colloidal science and technologies. Finally, we have made a number of our findings public through publication in peer-reviewed journals, specifically paper on colloids in Physical Review E and a significant review paper summarizing NASA-sponsored colloid work over the past two decades, in Annual Reviews of Condensed-Matter Physics, one of the most prestigious and important scientific journals.

	ED. NOTE (April 2015): Project continues with same title, grant #NNX13AQ48G, i.e., "Physics of Colloids in Space--NNX13AQ48G."
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Lu PJ, Weitz D. "Colloidal particles: Crystals, glasses, and gels." Annual Reviews of Condensed Matter Physics. 2013 Apr;4:217–33. http://dx.doi.org/10.1146/annurev-conmatphys-030212-184213 , Apr-2013
Articles in Peer-reviewed Journals	Nigro B, Grimaldi C, Ryser P, Varrato F, Foffi G, Lu PJ. "Enhanced tunneling conductivity induced by gelation of attractive colloids." Phys Rev E Stat Nonlin Soft Matter Phys. 2013 Jun;87(6):062312. PubMed PMID: 23848680 ; http://dx.doi.org/10.1103/PhysRevE.87.062312 , Jun-2013